

Home Assignment-week1day1

1) Classification Algorithm

a) Age <30 and Govt satisfaction less ,urban

Age-23.0 yrs

Govt. satisfaction-1.80 /10

Urban voter-Yes

Predicted preference-Party T

Party A-0.0%

Party T-99.6%

Party C-0.1%

Undecided-0.3%

Step 1 — each class gets its own score = base + (weight × this voter's value), added up

Party A

-11.4 pts ← base points

+ 0.038 × 23.0 (Age) = +0.866

+ 1.92 × 1.80 (Govt. satisfaction) = +3.46

- 1.06 × Yes (Urban voter) = -1.06

= -8.11 pts

Party T

11.4 pts ← base points

- 0.113 × 23.0 (Age) = -2.60

- 1.62 × 1.80 (Govt. satisfaction) = -2.91

+ 1.10 × Yes (Urban voter) = +1.10

= 6.95 pts

Party C

0.669 pts ← base points

$$+ 0.114 \times 23.0 \text{ (Age)} = +2.62$$

$$- 1.03 \times 1.80 \text{ (Govt. satisfaction)} = -1.85$$

$$- 1.33 \times \text{Yes (Urban voter)} = -1.33$$

$$= 0.103 \text{ pts}$$

Undecided

$$1.19 \text{ pts} \leftarrow \text{base points}$$

$$- 0.010 \times 23.0 \text{ (Age)} = -0.235$$

$$+ 0.231 \times 1.80 \text{ (Govt. satisfaction)} = +0.416$$

$$- 0.385 \times \text{Yes (Urban voter)} = -0.385$$

$$= 0.982 \text{ pts}$$

highest score wins $\rightarrow$ Party T

Step 2a — convert each score to e^{score} (always positive)

$$\text{Party A } e^{-8.11} = 0.000$$

$$\text{Party T } e^{6.95} = 1041$$

$$\text{Party C } e^{0.103} = 1.11$$

$$\text{Undecided } e^{0.982} = 2.67$$

$$\text{Total (add all 4 up)} = 1045$$

Step 2b — each class's share = its own $e^{\text{score}} \div$ that same total (1045)

$$\text{Party A } 0.000 \div 1045 = 0.0\%$$

$$\text{Party T } 1041 \div 1045 = 99.6\%$$

$$\text{Party C } 1.11 \div 1045 = 0.1\%$$

$$\text{Undecided } 2.67 \div 1045 = 0.3\%$$

All 4 shares add back up to 100% — that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

The trained model

Party A score = $-11.4 \text{ pts} \leftarrow \text{base points (starting score)} + 0.038 \times \text{Age} + 1.92 \times \text{Govt. satisfaction} - 1.06 \times \text{Urban voter}$

Party T score = $11.4 \text{ pts} \leftarrow \text{base points (starting score)} - 0.113 \times \text{Age} - 1.62 \times \text{Govt. satisfaction} + 1.10 \times \text{Urban voter}$

Party C score = $0.669 \text{ pts} \leftarrow \text{base points (starting score)} + 0.114 \times \text{Age} - 1.03 \times \text{Govt. satisfaction} - 1.33 \times \text{Urban voter}$

Undecided score = $1.19 \text{ pts} \leftarrow \text{base points (starting score)} - 0.010 \times \text{Age} + 0.231 \times \text{Govt. satisfaction} - 0.385 \times \text{Urban voter}$

Sample calculation — one real row

$45.7 \text{ yrs} \cdot 6.43 / 10 \cdot \text{not urban voter}$

Party A 2.70

Party T -4.21

Party C -0.742

Undecided 2.20

highest wins $\rightarrow$ Party A

actual Party A

b) Age >50 and Govt satisfaction less ,non urban

Make a prediction

Age-61.0 yrs

Govt. satisfaction-1.80 /10

Urban voter-No

Predict

Predicted preference-Party C

Party A-0.0%

Party T-1.4%

Party C-97.8%

Undecided-0.8%

Step 1 — each class gets its own score = base + (weight × this voter's value), added up

Party A

-11.4 pts ← base points

+ 0.038 × 61.0 (Age) = +2.30

+ 1.92 × 1.80 (Govt. satisfaction) = +3.46

– 1.06 × No (Urban voter) = +0

= -5.62 pts

Party T

11.4 pts ← base points

– 0.113 × 61.0 (Age) = -6.90

– 1.62 × 1.80 (Govt. satisfaction) = -2.91

+ 1.10 × No (Urban voter) = +0

= 1.55 pts

Party C

0.669 pts ← base points

+ 0.114 × 61.0 (Age) = +6.95

– 1.03 × 1.80 (Govt. satisfaction) = -1.85

– 1.33 × No (Urban voter) = +0

= 5.77 pts

Undecided

1.19 pts $\leftarrow$ base points

$- 0.010 \times 61.0$ (Age) = -0.623

$+ 0.231 \times 1.80$ (Govt. satisfaction) = +0.416

$- 0.385 \times \text{No}$ (Urban voter) = +0

= 0.978 pts

highest score wins $\rightarrow$ Party C

Step 2a — convert each score to e^{score} (always positive)

Party A $e^{-5.62} = 0.004$

Party T $e^{1.55} = 4.71$

Party C $e^{5.77} = 321$

Undecided $e^{0.978} = 2.66$

Total (add all 4 up) = 328

Step 2b — each class's share = its own e^{score} $\div$ that same total (328)

Party A $0.004 \div 328 = 0.0\%$

Party T $4.71 \div 328 = 1.4\%$

Party C $321 \div 328 = 97.8\%$

Undecided $2.66 \div 328 = 0.8\%$

All 4 shares add back up to 100% — that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

The trained model

Party A score = -11.4 pts $\leftarrow$ base points (starting score) + $0.038 \times \text{Age}$ + $1.92 \times \text{Govt. satisfaction}$ - $1.06 \times \text{Urban voter}$

Party T score = 11.4 pts $\leftarrow$ base points (starting score) - $0.113 \times \text{Age}$ - $1.62 \times \text{Govt. satisfaction}$ + $1.10 \times \text{Urban voter}$

Party C score = 0.669 pts ← base points (starting score) + 0.114×Age – 1.03×Govt. satisfaction – 1.33×Urban voter

Undecided score = 1.19 pts ← base points (starting score) – 0.010×Age + 0.231×Govt. satisfaction – 0.385×Urban voter

Sample calculation — one real row

45.7 yrs · 6.43 /10 · not urban voter

Party A 2.70

Party T -4.21

Party C -0.742

Undecided 2.20

highest wins → Party A

actual Party A

validate the percentage

2) Linear regression

a) QA Salary

Make a prediction

Experience-2.00 yrs

Skill rating-5.00 /10

Predict

Total package

11.7 LPA

Base salary — the starting value, before any feature counts

3.10 LPA

Experience

0.966 LPA per year × 2.00 yrs

= +1.93 LPA

Skill rating

1.34 LPA per point $\times$ 5.00 /10

= +6.71 LPA

TOTAL 11.7 LPA

The trained model

Salary_LPA in LPA =

3.10 LPA $\leftarrow$ base salary (minimum)

+ 0.966 LPA per year $\times$ Experience

+ 1.34 LPA per point $\times$ Skill rating

Sample calculation — one real row from the data

9.46 yrs $\cdot$ 5.78 /10

3.10 (base)

+ 0.966 $\times$ 9.46 = +9.14

+ 1.34 $\times$ 5.78 = +7.75

model says 20.0 LPA

actual value 19.4 LPA

b) Food Delivery time

Check the Calculation for price and time

Make a prediction

Distance-8.00 km

Traffic-4.20 /10

Predict

Total time

40.0 min

Base time — the starting value, before any feature counts

7.56 min

Distance

2.40 min per km × 8.00 km

= +19.2 min

Traffic

3.16 min per point × 4.20 /10

= +13.3 min

TOTAL 40.0 min

The trained model

Delivery_Minutes in min =

7.56 min ← base time (minimum)

+ 2.40 min per km × Distance

+ 3.16 min per point × Traffic

Sample calculation — one real row from the data

8.96 km · 8.46 /10

7.56 (base)

+ 2.40 × 8.96 = +21.5

+ 3.16 × 8.46 = +26.7

model says 55.8 min

actual value 55.4 min

Playground Link:

<https://testleaf-ai-playground.sudarshan210688.chatgpt.site/>